

Notes on Markov Decision Processes

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DISCOUNT FACTOR

The discount factor γ is a parameter in reinforcement learning that determines the importance of future rewards compared to immediate rewards. It is a value between 0 and 1, where: - A value of 0 means that the agent only cares about immediate rewards and ignores future rewards. We want to maximize the total discounted reward, which is given by the formula:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Where G_t is the total discounted reward at time step t , and R_{t+k+1} is the reward received at time step $t + k + 1$.

POLICY $\pi(s)$

A policy is a strategy that the agent follows to determine which action to take in each state. It is a mapping from states to actions, and is denoted by $\pi(a|s)$, which represents the probability of taking action a in state s .

stochastic policy: $\pi(a|s)$ is a probability distribution over actions given a state. It allows for randomness in action selection, which can be useful for exploration. deterministic policy: $\pi(s)$ is a function that maps each state to a specific action. It does not allow for randomness, and the agent will always take the same action in a given state.